

MAHDI PANAHI

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github.com/Mahpan2222 · [My Portfolio](#)

■ PROFILE

Final-year Computer Engineering student at Halmstad University. Co-author of an IEEE conference paper on cryptographic performance for V2X vehicle safety systems, accepted at IEEE ETFA 2026. Currently doing a paid industry thesis at Autoliv on real-time posture estimation with multi-IMU sensors. Alongside full-time studies, I have worked part-time for over three years, with references from managers at several levels.

■ PUBLICATIONS & RESEARCH

A Micro-benchmark of ECDSA Curves and Ed25519 for Low-Latency V2X Message Authentication

2025 – 2026

IEEE ETFA 2026 (Emerging Technologies & Factory Automation), Västerås, Sweden · Status: Accepted

- Co-authored with researchers at Halmstad University and Federal University of Rio Grande do Sul, Brazil
- Supervised by Edison Pignaton de Freitas (Halmstad University & UFRGS, Brazil), Wagner Ourique de Moraes and Alexandre dos Santos Roque
- Benchmarked three elliptic curve cryptographic algorithms (ECDSA secp256r1, brainpoolP256r1, Ed25519) for real-time V2X message authentication
- Key finding: Ed25519 achieves up to 27,000 signatures per second — fast enough for dense traffic within the strict 100 ms latency limit
- Implemented using Java Microbenchmark Harness (JMH) and the Bouncy Castle cryptography library
- Source code publicly available — github.com/obakha22

■ EDUCATION

B.Sc. Computer Engineering — Datateknik

2023 – present

Halmstad University (Högskolan i Halmstad), Halmstad

- Programme: Dataingenjör inom datateknik — covering embedded systems, sensor fusion, algorithms, computer networks and software development
- Final-year industry thesis in collaboration with Autoliv — see Degree Thesis below

■ DEGREE THESIS

Posture Estimation for Motorcycle Riders — Multi-IMU System

Feb 2025 – present

Autoliv, Sweden · Supervisor: Oscar Amador Molina

- Building a real-time posture estimation system using seven IMUs fitted into a motorcycle jacket and helmet
- Comparing Madgwick, Mahony and Extended Kalman Filter algorithms to handle sensor drift and keep orientation accurate over time
- Validating results against a video-based ground truth built with MediaPipe, measuring accuracy with RMSE of joint angles
- Responsible for the full hardware stack — ESP32 microcontrollers, microROS firmware and Wi-Fi data streaming
- Paid thesis position under a signed agreement with Autoliv

■ EXPERIENCE

Plant & Retail Associate

2023 – present

Blomsterlandet, Helsingborg · via Job and Talent

- Over three years of part-time work alongside full-time studies, at one of Sweden's largest garden and plant retail chains

- Day-to-day responsibilities include stock management, customer guidance and keeping the department in good order during busy periods
- Good working relationships across the team and with management — five managers have offered to act as references

Restaurant & Service Work

2019 – 2023

Multiple employers, Helsingborg area

- Worked summers from the age of 17 across various restaurant and service roles — learned to work under pressure and adapt quickly to new teams

Renovation & Painting Work (seasonal)

2020 – 2023

Private clients, Helsingborg area

- Took on interior painting and renovation jobs during summers, handling client contact and the work independently from start to finish

■ TECHNICAL SKILLS

Languages	Python · Java · C / C++ · PHP · JavaScript · Bash
Embedded	ESP32 · microROS · IMU integration · Wi-Fi data streaming
Algorithms	Extended Kalman Filter · Madgwick · Mahony · Sensor fusion
Machine Learning	Supervised learning · Classification · scikit-learn
Frontend	HTML · CSS · JavaScript
Databases	SQL · MySQL · Database design
Tools	MediaPipe · Git · ROS
Methods	Algorithm development · Data collection & validation · Technical documentation

■ LANGUAGES

Swedish	Fluent — professional working level
English	Fluent — professional working level
Farsi	Native

■ REFERENCES

Obey

Department Manager · Blomsterlandet

Phone: _____ | Email: _____

Josefin

Operations Manager · Blomsterlandet

Phone: _____ | Email: _____

Harald

Operations Manager · Blomsterlandet

Phone: _____ | Email: _____

Amina

Manager · Job and Talent

Phone: _____ | Email: _____

Christian Nils

Product Manager · Autoliv

Phone: _____ | Email: _____

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